

# THE HISTORY AND APPLICATIONS OF FOURIER ANALYSIS

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## INTRODUCTION TO FOURIER ANALYSIS

Fourier analysis is one of the most powerful mathematical tools for the study of the natural world and more. The applications of Fourier analysis are various and sundry, not to mention extremely useful; from quantum mechanics and electromagnetism to signal compression and even economics.

There are two main tools in Fourier analysis: the Fourier series and the Fourier transform. In a broad sense, the Fourier series decomposes functions into many, perhaps infinitely many, harmonic parts. In other words, the Fourier series of a given function is a sum of the component frequencies that make up that function. The Fourier transform, in general, takes a function in amplitude vs time and converts it to a function in amplitude vs frequency.

As mentioned previously, the applications of Fourier analysis are quite broad, and, as such, there are no specifically defined forms of the Fourier series and transform although they are all similar[1]. The form of each depends on the use case. For the sake of clarity in this paper, when speaking technically, we will use the form of the Fourier transform set forth in the *CRC Standard Mathematical Tables and Formulas*[2].

*Theorem 1:* If the integral exists, then Fourier transform of a function  $f(t)$  is:

$$(1) \quad \begin{aligned} \hat{f}(\omega) &= a \int_{-\infty}^{\infty} f(t) e^{\pm i t \omega} dt \\ f(t) &= b \int_{-\infty}^{\infty} \hat{f}(\omega) e^{\pm i t \omega} d\omega \end{aligned}$$

With the condition that  $ab = \frac{1}{2\pi}$  and that the exponent of the exponential must be opposite in the forward and reverse transforms.

We will also use the form of the Fourier series that is presented in that title.

*Theorem 2:* If  $f(x)$  is a bounded periodic function with a period of  $2L$  and  $f$  satisfies the Dirichlet conditions:

1.  $f(x)$  is continuous in any period except for a finite number of jump discontinuities.

2. In any given period,  $f(x)$  has finitely many maxima and minima.

Then the Fourier series of  $f(x)$ , written using a complex exponential is,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i w_n x},$$

(2) where  $w_n = \frac{n\pi}{L}$ , and

$$c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i w_n x} dx,$$

where  $n \in \mathbb{N}$ .

#### HISTORY OF FOURIER ANALYSIS

The history of Fourier analysis begins with Jean Baptiste Joseph Fourier, who introduced the first form of the Fourier transform, in 1822 in his famous book *The Analytical Theory of Heat* [3]. In this work Fourier, introduced the transform in two parts, the forward and inverse cosine Fourier transform:

$$(3) \quad \begin{aligned} F_c(u) &= \frac{2}{\pi} \int_0^{\infty} f(x) \cos(ux) dx \\ f(x) &= \int_0^{\infty} F_c(u) \cos(ux) du, \end{aligned}$$

and the forward and inverse sine Fourier transform:

$$(4) \quad \begin{aligned} F_s(u) &= \frac{2}{\pi} \int_0^{\infty} f(x) \sin(ux) dx \\ f(x) &= \int_0^{\infty} F_s(u) \sin(ux) du. \end{aligned}$$

These are not the familiar version of the Fourier transform used today. The first instance of a form of the Fourier transform that is used today was introduced in 1827 by Augustine Louis Cauchy [4]. Cauchy utilized Euler's formula to condense

(3) and (4) into:

$$(5) \quad \begin{aligned} F(u) &= \int_{-\infty}^{\infty} f(x)e^{-ixu}dx \\ f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u)e^{ixu}du, \end{aligned}$$

Under the condition that  $f$  is defined on the whole real line, is complex valued, and meets certain other conditions. From this form, many others have been adapted and derived. There is no definitive form, but they all follow the same basic principles.

The Fourier series was presented by Fourier in his famous work *Mémoire Sur la Propagation de la Chaleur Dans les Corps Solides*[5]. In this work he attempted to generate a trigonometric representation of all functions. He presented the cosine representation of the series as:

$$(6) \quad \begin{aligned} \varphi(y) &= a_1 \cos\left(\frac{\pi y}{2}\right) + a_3 \cos\left(3\frac{\pi y}{2}\right) + a_5 \cos\left(5\frac{\pi y}{2}\right) + \dots \\ a_n &= \int_{-1}^1 \varphi(y) \cos\left((2n+1)\frac{\pi y}{2}\right) dy. \end{aligned}$$

This was refined in by the work of Peter Gustav Lejeune Dirichlet, published in 1829, titled *Sur la Convergence des Séries Trigonométriques qui Servent à Représenter une Fonction Arbitraire Entre des Limites Données*[6]. In this work, he proved the convergence of the Fourier series of a function that satisfies the Dirichlet conditions, which were seen in Theorem 2. These two works, as well as the work of others such as Cauchy and Riemann led to the development of the modern Fourier series seen in Theorem 2.

It is worth noting that the formulae in this section were adapted from their original forms such that they may conform with modern mathematical notation styles.

## MAJOR THEOREMS AND RESULTS

Some major theorems and results pertaining to the Fourier transform and series are given in this section. For clarity's sake we will begin with those pertaining to the transform and move to those pertaining to the series.

**Major Theorems and Results: Fourier Transform.** Firstly, Theorem 1 is stated in the introduction since it and Theorem 2 are the focus of this paper, and it is clearly a major theorem pertaining to this subject.

*Riemann-Lebesgue lemma:* If  $f \in L^1(\mathbb{R})$ , then the Fourier transform of  $f$  is defined everywhere, uniformly continuous, and tends to zero as  $\omega$  tends to  $\pm\infty$ .

*Shannon's Sampling Theorem:* If  $f$  is a function band limited to  $[-\sigma, \sigma]$ , which is to say,

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\sigma}^{\sigma} F(\omega) e^{it\omega} d\omega,$$

Where  $F \in L^2(-\sigma, \sigma)$ , then  $f$  can be reconstructed from its sample values at,

$$t_k = \frac{k\pi}{\omega}, \text{ for } k \in \mathbb{Z},$$

by using the formula:

$$f(t) = \sum_{k=-\infty}^{\infty} f(t_k) \frac{\sin(\sigma(t - t_k))}{\sigma(t - t_k)}.$$

*Theorem 3:* The discrete Fourier transform of a sequence  $\{a_n\}_{n=0}^{N-1}$ , is a another sequence  $\{A_m\}_{m=0}^{N-1}$ , where.

$$A_m = \sum_{n=0}^{N-1} a_n e^{\frac{2\pi i m n}{N}}$$

*Properties:* Before mentioning famous results of the Fourier transform, we will discuss some of its properties. The Fourier transform turns convolution, differentiation, and integration into algebraic operations. This fact lead directly to several of its applications, such as solving differential equations.

The following three are some, not all, properties of the Fourier transform;

1.  $\widehat{f * g}(t) = \hat{f}(\omega) \times \hat{g}(\omega)$
2.  $\widehat{f^{(n)}}(\omega) = (-\omega i)^n \hat{f}(\omega)$
3.  $\int_{-\infty}^{\infty} \hat{f}(\omega) = \frac{1}{-\omega i} \hat{f}(\omega)$

*Fast Fourier Transform:* The fast Fourier transform is an efficient algorithm for calculating discrete Fourier transforms. This allows for a retinue of applications, that would otherwise be impractical. Like the Fourier transforms itself, the fast Fourier transform has no specific form, but rather it has several different forms suited to different applications.

The first fast Fourier transform was put forth by Cooley and Tukey in 1965 [7], and it reduced the complexity of a fast Fourier transform for  $O(n^2)$  to  $O(\frac{n}{2}\log_2(n))$ . This is done generally by splitting the set in question into several subsets and then computing the Fourier transform of each of those sets. There are many algorithms for performing fast Fourier transforms and their specific details are not relevant to this paper, but to see an overview of several algorithms, see [7].

**Major Theorems and Results: Fourier Series.** Firstly, Theorem 2 is stated in the introduction since it and Theorem 1 are the focus of this paper, and it is clearly a major theorem pertaining to this subject.

*Theorem 4:* The series presented in theorem 2 will converge to  $f(x)$  for every point where  $f(x)$  is continuous, and it will converge to the average of the left hand and right hand sided limits for jump discontinuity on  $f(x)$ .

*Theorem 5:* If two functions,  $f(x)$  and  $g(x)$ , have the same Fourier series representations, then  $f(x) = g(x)$ .

*Properties:* The Fourier series has properties similar to those of the Fourier transform. That is, convolution, differentiation, and integration become algebraic operations.

If  $g$  and  $p$  are suitable functions and  $f_{gn}$  and  $f_{pn}$  are their respective  $n$ th Fourier coefficients, then

1. The Fourier series of  $g(x) * p(x)$  would have coefficients  $T \times f_{gn} \times g_{pn}$
2. The Fourier series of  $\frac{d}{dx}f(x)$  would have coefficients  $i\omega_n f_{gn}$
3. The Fourier series of  $\int_{-\infty}^{\infty} g(x)dx$  would have coefficients  $\frac{f_{gn}}{i\omega_n}$ .

## APPLICATIONS OF FOURIER ANALYSIS

The applications of Fourier analysis are truly vast, and this paper cannot possibly touch on all of them. However, we will look to give an overview of several well known applications.

The Fourier transform is an extremely useful tool when solving differential equations. Recall property two of the Fourier transform. This turns the operation of differentiation into an algebraic operation, which, in turn, makes the computation of solutions for differential equations much less rigorous. For instance, consider the differential equation:

$$f''(t) + f'(t) + f(t) = g(x).$$

Applying property two and taking the Fourier transform of this gives,

$$\begin{aligned} -\omega^2 \hat{f}(\omega) - i\omega \hat{f}(\omega) + \hat{f}(\omega) &= \hat{g}(\omega) \\ \implies \frac{\hat{g}(\omega)}{-\omega^2 - i\omega + 1} &= \hat{f}(\omega) \end{aligned}$$

From this, we can simply use the inverse transform found in (1) to arrive at:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\hat{g}(\omega)e^{it\omega}}{-\omega^2 - i\omega + 1} d\omega$$

This is a simple example, but we can see how, on a more complex equation, this method could be extremely effective.

Fourier analysis is also extremely useful in signal and image processing. For example we could take a grey scale image and define each pixel in that image as a point  $(x,y)$  of the function  $f(t)$ . We could then apply a filter to this image by defining a filter function,  $g(t)$ , and then we would convolve the two functions. This filter function could be any that we desire. Perhaps  $g(t) = f(t) + C$ . However, this is computationally expensive. Herein lies the value of the Fourier transform as a computational tool. By applying property 1 of the Fourier transform, we can easily compute the convolution and then apply the inverse Fourier transform to arrive at

our desired result. Furthermore, we could use the fast Fourier transform in this case to lessen the computing power needed by an even greater degree.

We could also apply a filter to an image, or a signal, such as an audio input, by expanding its Fourier series and then applying a filter only to specific frequencies within the signal. We could then recover the new, filtered, function from the coefficients of that series and the result would be a filtered signal.

In several areas of physics, waves can be broken down using the Fourier series, with theorem two used to compute that series, and the component waves can then be analyzed. This is especially helpful when it comes to topics such as wave interference.

In general, Fourier analysis can best be viewed as a computational tool for applied mathematics. It allows complex problems to be greatly simplified and allows for insights into functions that would otherwise be infeasible.



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